



Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

Syllabus – C.1 Simple Harmonic Motion

C.1 word for word from the IB Physics guide, in its order. Tick a box when you can do it without your notes.

I. Guiding questions

- What makes the harmonic oscillator model applicable to a wide range of physical phenomena?
- Why must the defining equation of simple harmonic motion take the form it does?
- How can the energy and motion of an oscillation be analysed both graphically and algebraically?

II. Understandings

A. Standard level and higher level: 3 hours

Students should understand:

- ☐ 1. conditions that lead to simple harmonic motion
- ☐ 2. the defining equation of simple harmonic motion as given by $a = -\omega^2 x$
- ☐ 3. a particle undergoing simple harmonic motion can be described using time period T , frequency f , angular frequency ω , amplitude, equilibrium position, and displacement
- ☐ 4. the time period in terms of frequency of oscillation and angular frequency as given by $T = \frac{1}{f} = \frac{2\pi}{\omega}$
- ☐ 5. the time period of a mass–spring system as given by $T = 2\pi\sqrt{\frac{m}{k}}$
- ☐ 6. the time period of a simple pendulum as given by $T = 2\pi\sqrt{\frac{l}{g}}$
- ☐ 7. a qualitative approach to energy changes during one cycle of an oscillation.

Higher level only

B. Additional higher level: 4 hours

Students should understand:

- ☐ 1. that a particle undergoing simple harmonic motion can be described using phase angle
- ☐ 2. that problems can be solved using the equations for simple harmonic motion as given by
 - ☐ a. $x = x_0 \sin(\omega t + \phi)$
 - ☐ b. $v = \omega x_0 \cos(\omega t + \phi)$
 - ☐ c. $v = \pm \omega \sqrt{x_0^2 - x^2}$
 - ☐ d. $E_T = \frac{1}{2} m \omega^2 x_0^2$
 - ☐ e. $E_p = \frac{1}{2} m \omega^2 x^2$.

III Guidance and IV Linking questions are on page 2.

Syllabus – C.1 Simple Harmonic Motion

C. Wave behaviour · Standard level and higher level: 3 hours · Additional higher level: 4 hours

III. Guidance

- A. The significance of the minus sign in the defining equation for simple harmonic motion should be understood.
- B. Energy changes during simple harmonic motion (kinetic, potential and total) should be described qualitatively.
- C. A quantitative approach to energy changes during simple harmonic motion is required at higher level only.
- D. Radians are used for phase angle calculations.

IV. Linking questions

- A. How can greenhouse gases be modelled as simple harmonic oscillators? → B.2 Greenhouse effect
- B. How can circular motion be used to visualize simple harmonic motion? → A.2 Forces and momentum
- C. How does damping affect periodic motion? → C.4 Standing waves and resonance
- D. How can the understanding of simple harmonic motion apply to the wave model? (NOS) → C.2 Wave model
- E. What physical explanation leads to the enhanced greenhouse effect? (NOS) → B.2 Greenhouse effect

Source: IB Diploma Programme *Physics guide*, first assessment 2025, pp. 45–46. Text reproduced verbatim. The guide uses bullets only: the outline labels (I, A, 1, a) and the arrows to linked topics are added.